

Sakchham Shrestha

sakchham.shrestha99@gmail.com | (469) 844-0159 | [LinkedIn](#)

PROFESSIONAL SUMMARY:

Results-driven **Senior Data Engineer** with **7+ years** of experience architecting and scaling cloud-native data platforms and lakehouse ecosystems across **healthcare, finance, and insurance**. Proven expertise designing enterprise **batch, real-time, and event-driven architectures** on **Azure, AWS, and GCP** in **compliance-heavy environments** (HIPAA, SOX, 21 CFR Part 11).

Strong background in **distributed data processing, Medallion architecture, data governance, and Infrastructure-as-Code** automation. Skilled in **Python, PySpark, SQL, Airflow, dbt, Terraform, and CI/CD frameworks**. Demonstrated leadership in optimizing performance, reducing cloud costs, and mentoring engineering teams while building secure, production-grade data systems.

TECHNICAL SKILLS:

- **Cloud Platforms & Services:** Azure (ADLS Gen2, Synapse, Databricks, Data Factory, Purview, Bicep, Event Hubs, Azure DevOps), AWS (S3, Redshift, Lambda, Glue, RDS, EMR, EKS, IAM, Lake Formation, MSK, CloudWatch), GCP (BigQuery, Cloud Storage, Pub/Sub)
- **Programming & Scripting:** Python, Scala, Java, R, SQL (T-SQL, PL/SQL), Pandas, NumPy, JavaScript
- **Big Data Engineering & Orchestration:** Apache Spark (PySpark, Spark SQL, Flink), Databricks, Airflow, dbt, Terraform, CI/CD Pipelines, ETL/ELT Automation, Data Integration
- **Data Architecture:** Medallion Architecture, Data Lakehouse, Data Lake, Data Modeling, Schema Evolution, Data Contracts, Star Schema, Snowflake Schema
- **Databases & Storage Systems:** PostgreSQL, MySQL, Oracle, SQL Server, MongoDB, DynamoDB, Cassandra, HBase, NoSQL, Delta Lake, OLAP, OLTP
- **Streaming & Messaging Platforms:** Apache Kafka, AWS Kinesis, AWS MSK, Azure Event Hubs, RabbitMQ
- **Infrastructure & DevOps:** Docker, Kubernetes (EKS, AKS), Terraform, AWS CloudFormation, Jenkins, **Git**, GitHub Actions, Helm, Prometheus, CI / CD Pipelines
- **Governance & Observability:** Great Expectations, Datadog, SLA/SLO Data Reliability, Lineage Tracking
- **Analytics & BI Tools:** Power BI, Tableau, Snowflake, Redshift Spectrum, Looker
- **Machine Learning & Analytics:** RAG pipeline, SageMaker, Scikit-learn, TensorFlow, PyTorch, MLflow, Predictive Modeling, Time Series Forecasting, Model Deployment
- **API & Integration Technologies:** RESTful APIs, GraphQL, Flask, FastAPI, TensorFlow, OpenAPI Specification
- **Security & Compliance:** SOX, HIPAA, GDPR, 21 CFR Part 11, PII, GxP, RBAC, Encryption at Rest/In-Transit
- **Project Delivery & Agile Methods:** Agile (SCRUM), Jira, SDLC, Confluence, ServiceNow, Backlog Grooming
- **Data Formats:** JSON, CSV, Parquet, Excel, XML, YAML

PROFESSIONAL EXPERIENCE:

CVS Health | Woonsocket, RI

Oct 2023 - Present

Senior Data Engineer

- Standardized SQL transformations with **dbt**, boosting reliability and metadata visibility across 100+ data models, which improved governance and reduced errors in downstream reporting.
- Automated orchestration with **Airflow** and **Step Functions**, reducing manual intervention by 40% and strengthening **SLA compliance** with faster recovery during failures.
- Designed reusable **ETL** workflows in **ADF** and architected a **Medallion Architecture** (Bronze/Silver/Gold) on **ADLS Gen2**, enabling **schema** evolution, lifecycle tracking, and scalable ingestion for structured/semi-structured healthcare data.
- Modeled and optimized **Star/Snowflake schemas** in **Synapse**, improving query performance by 40% and powering actuarial, **compliance**, and dashboards with self-service access via **Power BI** and **Synapse Studio**.

- Deployed secure pipelines and API-based **microservices** using **Azure Key Vault, RBAC, Private Endpoints**, and **VNET Integration**, enabling real-time healthcare data sharing across 50+ partner integrations while ensuring **HIPAA and SOC2 compliance**.
- Integrated **CI/CD** pipelines using **Azure DevOps**, automating deployments of ADF, Databricks notebooks, and Synapse artifacts using **YAML** and **Terraform**.
- Monitored pipeline health with **Azure Monitor, Log Analytics**, and **CloudWatch/SNS alerts**, achieving more than 98% availability and reducing incident response by 25%.
- Implemented **end-to-end data governance** and **lineage** using **Azure Purview, OpenLineage**, and **Glue Catalog**, integrating **ADF, Databricks**, and **Synapse** to support audit readiness, **HIPAA/GDPR compliance**, and fine-grained **data cataloging** with **PII tagging**.
- Developed scalable **PySpark** pipelines in **Azure Databricks** with **notebook-driven** frameworks for complex data cleansing, enrichment, **schema drift detection**, data quality enforcement, and streaming window aggregation across distributed sources.
- Engineered real-time ingestion with **Azure Event Hubs, Stream Analytics**, and Spark Structured Streaming, and integrated batch/streaming into **Azure Event-Driven Architecture** to enable low-latency routing and resilient, decoupled data services.
- Provisioned secure infrastructure with **Terraform** and **ARM Templates**, automating **Glue jobs, clusters**, and **IAM roles** to cut deployment cycles from days to hours, reducing human errors.
- Led migration of **OLAP/OLTP** systems from **SQL Server, Oracle**, and **MongoDB** into **Azure Data Lake**, improving scalability and eliminating \$250K/year in on-prem costs.
- Built **RAG pipelines** in Azure Databricks to generate vector embeddings from healthcare claims and clinical notes, enabling semantic search and LLM-assisted analytics.
- Mentored junior engineers on **data security, CI/CD**, and **Azure governance**, while leading **cross-functional** design reviews to improve delivery speed and establish best practices for scalable, production-grade data pipelines.

Technologies Used: Azure (ADF, ADLS Gen2, Synapse Analytics, Synapse Studio, Azure Databricks, Event Hubs, Stream Analytics, Azure Monitor, Log Analytics, Azure Purview, Azure DevOps, RBAC, VNET Integration), AWS (Glue Data Catalog, CloudWatch, SNS), dbt, Apache Airflow, Azure Step Functions, PySpark, Spark Structured Streaming, Power BI, OpenLineage, Terraform, ARM Templates, SQL Server, Oracle, MongoDB, YAML, Medallion Architecture (Bronze/Silver/Gold)

Johnson & Johnson | Santa Clara, CA

Jan 2021 - Aug 2023

Data Engineer

- Designed and scaled **streaming data platform** on AWS Kinesis processing **500K+ events/second**, supporting real-time feature engineering for ML models with **<100ms latency** SLA compliance.
- Built **automated feature pipelines** integrating real-time Kinesis streams with batch Redshift datasets for SageMaker model training, reducing feature engineering cycle time by **50%**.
- Implemented **Lake Formation governance framework** with column-level security policies, enabling secure data sharing across **12 business units** while maintaining HIPAA and SOX compliance controls.
- Optimized **AWS Glue ETL jobs** through job bookmarks, partition pruning, and resource allocation tuning, reducing execution time by **55%** and monthly AWS bills by **\$40K**.
- Developed **data cataloging strategy** using Glue Data Catalog and custom metadata tagging, improving data discoverability for **300+ analysts** and reducing time-to-insight by **3 days**.
- Containerized **ECS Fargate ETL components** using Docker, implementing **auto-scaling policies** and reducing infrastructure costs by **30%** while improving fault tolerance and deployment frequency.
- Established **CI/CD pipelines** with AWS CodeBuild/CodePipeline and GitHub Actions, enabling **20+ weekly deployments** with zero-downtime updates and automated rollback capabilities.
- Created **data quality frameworks** using Great Expectations, validating **100+ daily pipelines** with automated remediation and stakeholder notifications for SLA breaches.
- Designed **cost optimization strategy** reducing AWS spending by **28%** through reserved instances, spot pricing, and rightsizing recommendations without compromising SLA targets.

Technologies Used: AWS (S3, Glue, Redshift, Lambda, Kinesis, MWAA, Lake Formation, CodePipeline, SageMaker, Athena), Python, PySpark, Terraform, Docker, SNS, SQS, Vault, Lambda, Kinesis, CodePipeline, ECS Fargate, PySpark, Terraform, Docker, Great Expectations

JPMorgan Chase | Plano, TX

Jan 2019 – Dec 2020

ETL Developer

- Developed and maintained **200+ SSIS ETL workflows** ingesting structured and unstructured data from financial systems, processing **20M+ records daily** with **99.7% accuracy** for actuarial and underwriting divisions.
- Designed **dimensional data warehouse** using Kimball methodology with star/snowflake schemas, supporting **50+ SSRS reports** and reducing report generation time by **65%**.
- Built **T-SQL stored procedures and views** implementing complex business logic for data transformations, achieving consistent **sub-second query performance** for analytics and compliance reporting.
- Optimized **SQL Server database performance** through indexing strategies, query tuning, and partition design, improving batch load times from **4 hours to 45 minutes** for nightly EOD processes.
- Established **data quality validation framework** with automated reconciliation controls, reducing post-load data anomalies by **92%** and eliminating manual audit exceptions.
- Automated **SQL Agent jobs and batch scheduling** across dev/test/prod, reducing manual data load interventions by **85%** and improving operational efficiency for 15-person analytics team.
- Led **legacy system migration** from Informatica PowerCenter to SSIS, reducing licensing costs by **\$200K annually** while maintaining 100% feature parity and zero data loss.
- Managed **ODS and data marts** supporting downstream BI consumption, implementing slowly changing dimensions and audit trails to meet regulatory requirements for financial services.
- Created **data lineage documentation and metadata repositories** using Confluence and Visio, improving data transparency for **100+ business stakeholders** and reducing data discovery time by **40%**.
- Implemented **version control for ETL packages** using TFS, establishing change tracking protocols and enabling safer deployments with automated rollback procedures.

Technologies Used: SQL Server, SSIS, SSRS, T-SQL, Crystal Reports, Informatica PowerCenter, Query Optimization, SQL Agent, TFS, Visio, Dimensional Modeling, OLAP/OLTP, Confluence

EDUCATION:

Master of Science in Cybersecurity

Southeast Missouri State University - Cape Girardeau, Missouri